

Lucia Raciti

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EDUCATION

University of Notre Dame Notre Dame, IN	May 2027
<i>Bachelor of Science</i>	<i>GPA: 3.9</i>
Major: Computer Science	
Study Abroad: Dublin, Ireland	May-June 2024
Saint Mary's College Notre Dame, IN	May 2026
<i>Bachelor of Science</i>	<i>GPA: 3.96</i>
Major: Mathematics	

TECHNICAL SKILLS

Programming: Python, C, R, MATLAB, Excel – applied in data analysis, machine learning, and visual models
Data Analysis: Skilled in statistical modeling, multivariate calculus, and visualization for complex datasets

WORK AND PROJECT EXPERIENCE

Atlantic IPS Dublin, Ireland	May 2025 – Present
<i>Intern</i>	
- Designed and tested AI-driven prompts to streamline patent classification and infringement searches, enhancing the precision of intellectual property analysis - Optimized AI resource utilization by collaborating with cross-functional teams, improving efficiency and accuracy in patent-related workflows	
Mathematical Research Saint Mary's College	January 2024 – June 2025
<i>Research Assistant</i>	
- Analyzed 500+ data points on woodpecker populations, applying six different machine learning algorithms in Python to identify correlations between climate patterns and species decline - Optimized predictive accuracy by refining model selection, contributing key insights to conservation research - Presented findings at the Join Mathematics Meetings (JMM) Conference, engaging with 30+ researchers to discuss biodiversity trends and the adaptations occurring because of climate change	
Mathematical Contest in Modeling International Competition	January 2025
<i>Team Member</i>	
- Developed a mathematical model integrating differential equations, statistical analysis, and optimization techniques to assess sustainable tourism practices - Conducted an in-depth literature review and data analysis, refining assumptions to improve predictive outcomes	
Sustainable Farm Market Analysis Saint Mary's College	Dec 2024
<i>Team Member</i>	
- Applied R-based statistical modeling and hypothesis testing to evaluate market accessibility and community impact - Identified key consumer behavior trends through data visualization, changing complex datasets into clear insights - Delivered data-driven recommendations to stakeholders, leading to more targeted outreach initiatives	

LEADERSHIP AND ACTIVITIES

Pi Mu Epsilon	May 2023-Present
<i>President</i>	
- Led chapter meetings and organized events promoting excellence in mathematics and problem-solving - Collaborated with faculty and peers to increase student engagement in mathematical research and awareness of professional development opportunities within the field	
Mathematics and Computer Science Club	May 2022-Present
<i>Secretary</i>	
- Managed club communications, meeting agendas, and event documentation - Supported executive team initiatives to enhance student participation and promote STEM collaboration	